An Infinite-Width Analysis on the Jacobian-Regularised Training of a Neural Network

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Jacobian Regularisation

To increase the robustness of NN, [Hoffman et al., 2019] suggested regularising Jacobian of NN as

$$\mathcal{L}(\theta) := \frac{1}{2N}\sum_{i=1}^{N} \ell\left(f(x^{(i)};\theta), y^{(i)}\right) + \lambda \|J(f^{(i)})\|^2$$

for $0 < \lambda \leq 1$ where $J(f)$ is the Jacobian of f defined as

$$J(f)(x) := \left(\frac{\partial f(x)}{\partial x_0}, \dots, \frac{\partial f(x)}{\partial x_{d_0-1}}\right)^{\intercal}.$$

By including Jacobian regularisation, the predictor becomes flat near the training data.

Neural Tangent Kernel

Neural Tangent Kernel [Jacot et al., 2018] is a theoretical framework for analysing optimisation, generalisation, and inductive bias of NN. The object we are concerned with is

$$\overline{\Theta}_\theta(x, x') = \left\langle \frac{\partial f(x;\theta)}{\partial \theta}, \frac{\partial f(x';\theta)}{\partial \theta} \right\rangle.$$

Several major results include

- At initialisation, the neural network converges to the Gaussian process under an infinite width limit.
- At initialisation, the **neural tangent kernel** converges to deterministic kernel.
- During training, the **neural tangent kernel** stays close to its value at initialisation.
- Sufficiently wide neural networks **globally converges**, i.e., the training loss goes to zero.
- The learned predictor function can be expressed as **kernel regressor**.

Assumptions

We consider MLP without bias, defined as

$$h^{(1)}(x) := \phi(W^{(1)}x),$$

$$h^{(l)}(x) := \phi\left(\frac{1}{\sqrt{d_{l-1}}} W^{(l)} h^{(l-1)}(x)\right),$$

$$f(x) := \frac{\kappa}{\sqrt{d_L}} W^{(L+1)} h^{(L)}(x).$$

We assume that

- The loss is a mean squared error $\ell(\hat{y}, y) = (\hat{y} - y)^2$.
- The weights are initialised by i.i.d. Gaussian $(W^{(l)})_{ij} \sim \mathcal{N}(0,1)$.
- The width of hidden layers are all the same, i.e., $d_1 = \cdots = d_L = 1$.
- The output of the network is scaled by small enough $\kappa > 0$.
- The activation ϕ, and its first and second derivatives are Lipschitz continuous.
- The activation ϕ is normalised such that $\mathbb{E}_{z\sim\mathcal{N}(0,1)}[\phi(z)^2] = 1$.
- The inputs have the unit norm, $\|x^{(i)}\| = 1$ for all $i = 1, \dots, N$.
- The outputs are bounded, $|y^{(i)}| \leq 1$ for all $i = 1, \dots, N$.

Jacobian Neural Tangent Kernel

We adapt the NTK to the Jacobian regularisation setting JNTK $\Theta_\theta : \mathbb{R}^{d_0} \times \mathbb{R}^{d_0} \to \mathbb{R}^{(1+d_0)\times(1+d_0)}$ is defined as follows: for $\alpha, \beta = 1, \dots, d_0$,

$$\Theta_\theta(x, x')_{00} := \left\langle \frac{\partial f(x;\theta)}{\partial \theta}, \frac{\partial f(x';\theta)}{\partial \theta} \right\rangle,$$

$$\Theta_\theta(x, x')_{\alpha 0} := \left\langle \frac{\partial^2 f(x;\theta)}{\partial \theta\, \partial x_\alpha}, \frac{\partial f(x';\theta)}{\partial \theta} \right\rangle,$$

$$\Theta_\theta(x, x')_{0\beta} := \left\langle \frac{\partial f(x;\theta)}{\partial \theta}, \frac{\partial^2 f(x';\theta)}{\partial \theta\, \partial x'_\beta} \right\rangle,$$

$$\Theta_\theta(x, x')_{\alpha\beta} := \left\langle \frac{\partial^2 f(x;\theta)}{\partial \theta\, \partial x_\alpha}, \frac{\partial^2 f(x';\theta)}{\partial \theta\, \partial x'_\beta} \right\rangle$$

where $\theta = \{W^{(l)}\}_{l=1}^{L+1}$, and the inner product is entry-wise inner product $\mathbb{R}^{d_0 d_1 + \cdots + d_L d_{L-1}}$.

NNGP Convergence (Informal)

At initialisation, the neural network and its Jacobian converges weakly in finite marginal to $(1+d_0)$-dimensional zero-mean Gaussian Process with covariance kernel $\Sigma : \mathbb{R}^{d_0} \times \mathbb{R}^{d_0} \to \mathbb{R}^{(1+d_0)\times(1+d_0)}$ which admits the following form:

$$\Sigma(x, x')_{00} = \overline{\Sigma}(x, x'), \qquad \Sigma(x, x')_{\alpha 0} - \frac{\partial}{\partial x_\alpha}\Sigma(x, x')_{00},$$

$$\Sigma(x, x')_{0\beta} - \frac{\partial}{\partial x'_\beta}\Sigma(x, x')_{00}, \qquad \Sigma(x, x')_{\alpha\beta} = \frac{\partial^2}{\partial x_\alpha \partial x'_\beta}\Sigma(x, x')_{00},$$

where $\overline{\Sigma}(x, x')$ is the *standard* NNGP kernel.

JNTK at Initialisation (Informal)

At initialisation, the JNTK Θ_{θ_0} converges almost surely in finite marginal to its deterministic counterpart Θ, which admits following form:

$$\Theta(x, x')_{00} = \Theta(x, x'), \qquad \overline{\Theta}(x, x')_{\alpha 0} - \frac{\partial}{\partial x_\alpha}\Theta(x, x')_{00},$$

$$\Theta(x, x')_{0\beta} - \frac{\partial}{\partial x'_\beta}\Theta(x, x')_{00}, \qquad \Theta(x, x')_{\alpha\beta} = \frac{\partial^2}{\partial x_\alpha \partial x'_\beta}\Theta(x, x')_{00},$$

where $\overline{\Theta}$ is the *standard* NTK.

Assumption (JNTK Gram Matrix is Positive Definite)

Construct the Gram matrix of JNTK $\Theta(x^{(1:N)}, x^{(1:N)}) \in \mathbb{R}^{N(1+d_0)\times N(1+d_0)}$ by stacking $\Theta(x^{(i)}, x^{(j)}) \in \mathbb{R}^{(1+d_0)\times(1+d_0)}$. We assume that the JNTK gram matrix is positive definite, i.e., the smallest eigenvalue is nonzero:

$$\lambda_{\min}\left(\Theta(x^{(1:N)}, x^{(1:N)})\right) = \lambda_0 > 0.$$

Theorem (Constancy of JNTK and Training Dynamics)

Assume that all the assumptions hold. Let θ_t be the parameter of MLP at time t trained with gradient descent with learning rate η. Write $x^{(1:N)}$ as the training data, and x^* with $\|x^*\| = 1$ the test point we are concerned. Then there exists a function F, G, H such that if the following three conditions hold:

(Network is large enough) $\quad d \geq F(x^{(1:N)}, x^*, L, \epsilon, \delta, \lambda)$

(Scaling is small enough) $\quad \kappa \leq G(x^{(1:N)}, x^*, L, \epsilon, \delta, \lambda)$

(Learning rate is small enough) $\quad \eta \leq \leq H(x^{(1:N)}, x^*, L, \epsilon, \delta, \lambda)$

then followings hold with probability $\geq 1 - \delta$:

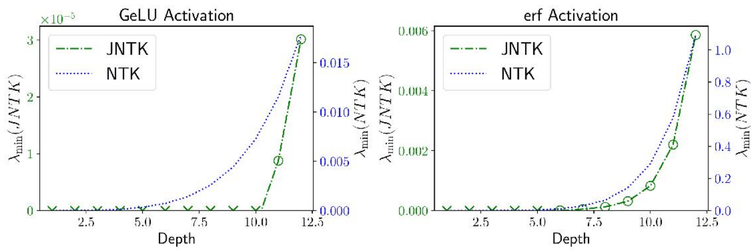
(Lazy training happens) $\quad \left\|W_0^{(l)} - W_t^{(l)}\right\|_p \leq O(\sqrt{d}\log d), \quad (p \in \{1, 2, \infty\})$

(JNTK stays close to its initialisation) $\quad \left\|\Theta_{\theta_0}(x^{(1:N)}, x^{(1:N)}) - \Theta_{\theta_t}(x^{(1:N)}, x^{(1:N)})\right\|_F \leq \epsilon$

(Network linearly converges) $\quad \mathcal{L}(\theta_t) \leq \left(1 - \frac{\eta\kappa^2\lambda_0}{4N}\right)^k \mathcal{L}(\theta_0)$

(Network is kernel regressor) $\quad |f_{\theta_\infty}(x^*) - f_\Theta(x^*)| \leq \epsilon.$

Certification of Assumption



There exists activation ϕ and dataset $x^{(1:N)}$ such that the standard NTK gram matrix $\Theta(x^{(1:N)}, x^{(1:N)})_{00}$ is positive definite, but JNTK gram matrix is not.

Validation of Theory

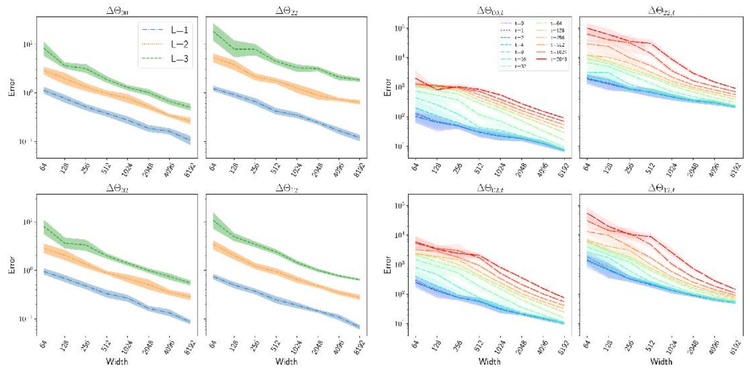
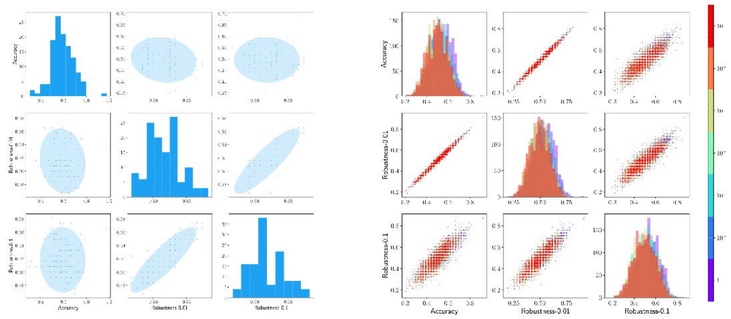


Figure 1. (Left) The $\|\cdot\|_{\infty,\infty}$ difference of NTK at its initialisation and limiting counterpart. (Right) $\|\cdot\|_{\infty,\infty}$ difference of NTK at during training and limiting counterpart.

Eigenanalysis of JNTK



Akin to [Tsilivis and Kempe, 2022], consider the eigendecomposition $(\lambda_i, v_i)_{i=1}^{N \text{ or } N(1+d_0)}$ of standard NTK or JNTK which gives eigenfeature

$$f_i(x^*) = \Theta(x^*, x^{(1:N)})\lambda_i^{-1} v_i v_i^{\intercal} y^{(1:N)}.$$

We plot the test accuracy and test robustness against L_2 single-step adversarial attack with thresholds 0.01, 0.1.

Without Jacobian regularisation, the test accuracy and robustness show less or no correlation. With Jacobian regularisation, we can see a strong correlation, suggesting that Jacobian regularisation promotes alignment of *useful* feature and *robust* feature.

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